

Blockchain Technology: Everything You Need to Know About It

Blockchain technology has gained significant attention within a short span of time, and is predicted to grow exponentially and revolutionise the current financial ecosystem. This article takes a quick look at the foundational concepts of blockchain technology.



In the existing financial setup, banks and governments act as intermediaries for any electronic payment (by an individual or business), creating trust and certainty by facilitating the transaction of goods and services. However, they often cause delays and impede the free flow of business due to the time it takes to process transactions and meet regulatory requirements. And even a system managed by a central governing authority is prone to being hacked or corrupted.

Therefore, there is need for a technology that enables direct business-to-business (B2B) or peer-to-peer transactions without the involvement of any banks, government or intermediary third parties. In 2008, blockchain technology was invented by Satoshi

Nakamoto (a pseudonym for a person or group) to address these issues. This technology creates a decentralised digital public record of transactions, which is secure, anonymous, tamper-proof and unchangeable. In this technology, the block is a part of a blockchain, which is a combination of three items — a hash pointer to the previous block, a timestamp, and transaction data such as the sender, receiver and number of Bitcoins.

Benefits of blockchain compared to banks or other intermediary systems

- Unlike intermediary systems that maintain private databases of records, a blockchain makes all records public with increased transparency.

- Blockchain technology eliminates expensive intermediary fees, thereby removing a substantial burden on individuals and businesses.
- Hacking attacks and fraudulent activities that typically impact large central intermediaries are not possible with blockchain technology.

Therefore, security and transparency are the two main features and advantages of blockchain technology. And the fact that a blockchain is immutable, makes tampering with information practically impossible.

Role of cryptocurrency in blockchain

Cryptocurrency is a combination of cryptography and currency. It is digital currency that uses encryption techniques to regulate the generation of monetary units and verify the transfer of funds, independent of any central authority. These currencies are not issued by a central government and are used to facilitate the transfer of funds between two parties.

Bitcoin, Litecoin, Ripple, Ethereum and Zcash are the popular cryptocurrencies currently. Among these, Bitcoin is the world's foremost, invented by the same person (or group) Satoshi Nakamoto in 2009. One of the leading official sites that monitors the trading of digital cryptocurrencies is CoinCap.

